DE PI

**DEPI | Digital Egypt Pioneers Initiative**
Track: Data Engineering

# Web-to-Warehouse Product Intelligence Pipeline

Compliance-first ETL | Kimball warehouse | Data-quality gate | REST API | React dashboard

| 35 | 29/23 | 12x6 | 219 | 575 | 592 |
|---|---|---|---|---|---|
| ingestion sources | tables / views | DQ rules x dimensions | REST operations | catalogued features | automated tests |

## 01 Overview
Web-to-Warehouse product intelligence

- permitted web sources
- retailer catalog

## 02 Problem
Manual collection is slow and inconsistent

- stale spreadsheets
- no history

## 03 Solution
Compliance-first ETL pipeline

- robots.txt aware
- deduplicated

## 04 Architecture
Airflow orchestrates 9 stages

- ingest, clean, load
- detect, reconcile

## 05 Warehouse
29 tables, 23 analytical views

- Alembic-managed schema
- PostgreSQL and MySQL twins

## 06 Data Quality
12 rules across 6 dimensions

- score persisted per run
- 98.26 measured

## 07 Forecasting
Prices, anomalies, advice

- damped Holt-Winters
- MAD, sigma and IQR detectors

## 08 Operations
219 REST operations, 23 routers

- JWT, TOTP and scoped keys
- leased jobs and SSE

## 09 Dashboard
21 screens, 5 themes, 4 report formats

- command palette, installable PWA
- HTML, JSON, CSV and PDF

## 10 Evidence
592 tests, 104 smoke checks, CI green

- mypy and Ruff clean
- 33 documents, 83 diagrams

## TECH STACK

Python 3.12 · FastAPI · Airflow 2.10 · SQLAlchemy · NumPy · Alembic · WeasyPrint · React 19 · TypeScript · Tailwind · Recharts · PostgreSQL 16 · MySQL 8.4 · Argon2id + TOTP · Server-Sent Events · Docker Compose · GitHub Actions · pytest + mypy + Ruff

## DEVELOPMENT TIMELINE

Plan → Design → Build → Test → Deploy → Maintain

## PROJECT OUTCOMES

- Working pipeline
- 29-table warehouse, 23 views
- 12-rule DQ gate
- REST API, 219 operations
- 592 automated tests
- 104 API smoke checks
- React dashboard, 21 screens
- Alembic-managed schema
- 33 documents, 83 diagrams
- PDF and CSV reports



github.com/Ahmedbakr78/Web-to-Warehouse-Product-Intelligence-Pipeline